Immunity and vaccination

9C Threat from disease · Lesson 2 of 7

Date _____ Class _____

Today I can

- ☐ I can describe how phagocytes and lymphocytes destroy microorganisms in the body.
- ☐ I can explain why you only get many infectious diseases once, and how vaccination protects people and communities.

What happens today

1. Do Now: once is enough
2. Learn: Two kinds of white blood cell · Why you are ill before you get better · Memory cells make you immune · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 39 to 41 · Workbook pages 29 to 30

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
white blood cell	a blood cell that fights pathogens
phagocyte	a white blood cell that surrounds, ingests and digests microorganisms
lymphocyte	a white blood cell that makes antibodies
antibody	a protein that sticks to one type of pathogen
specific	fits only one type of pathogen
memory cell	a lymphocyte that stays in the body ready to make antibodies quickly
immune	protected against a disease, so you do not become ill with it
vaccine	a substance containing parts of a pathogen, given to make a person immune
virus	a very small pathogen that can only copy itself inside a cell
dose	the amount of a medicine given at one time
half-life	the time for the amount of a medicine in the body to fall by half

Quick Review

once is enough

Many people had chickenpox as a small child. If a friend has it now, they will almost certainly not catch it again.

1 Suggest why you only get chickenpox once.

2 What is a pathogen? (last lesson)

3 Name two things your body does to fight an infection.

Core 1 of 2

Everyone does these. Answer on the card or under it.

1 Complete the text using words from the word bank. Some words are used more than once.

When you are infected by a _____, white blood cells start to attack.

_____ ingest some of the microorganisms, while _____ start to produce

_____ that stick to the outside of the microorganisms. These proteins may cause the

_____ to burst open, or cause several microorganisms to _____ together,

which makes it easier for them to be _____.

2 Match each key word to its meaning. Draw a line.

phagocyte	a lymphocyte that stays in the body, ready to make antibodies quickly
lymphocyte	a protein that sticks to one type of pathogen
antibody	a white blood cell that ingests and digests microorganisms
memory cell	contains parts of a pathogen and makes a person immune
vaccine	a white blood cell that makes antibodies

Core 2 of 2

Everyone does these. Answer on the card or under it.

3 Tick one box in each row to show whether the statement is true or false.

Statement	True	False
You are ill before you get better because new antibodies take days to make.		
One type of antibody fits every type of pathogen.		
A vaccine gives you a mild case of the disease.		
Memory cells make the second response faster and larger.		
Vaccinating most people can stop a disease spreading in a community.		

Challenge 1 of 3

When Core is finished. Full sentences.

C1 A vet uses an antibiotic called ceftiofur. The table gives the dose and the half-life for some animals. The half-life is the time for the amount of antibiotic in the animal's body to fall by half.

Species	Dose (mg of antibiotic per kg of animal)	Half-life (hours)
alpaca	6.6	45
American black duck	6.6	32
ball python	10	64
goat	15	37

a) State what is meant by the dose of a medicine.

Challenge 2 of 3

When Core is finished. Full sentences.

C1 *continued*

b) To be effective, the concentration of ceftiofur in a ball python must stay above 2.5 mg/kg. Calculate how many days after the first dose a second dose is needed. Show your working.

c) Calculate the dose, in mg, for a 15 kg goat.

Challenge 3 of 3

When Core is finished. Full sentences.

C2 Explain how vaccinating most of the people in a community protects people who cannot be vaccinated, such as newborn babies.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Name the two types of white blood cell and say what each one does.

2 Why do you only get measles once?

3 What does a vaccine contain, and why does it not give you the disease?

Today I can: Goal 1 ☐ Goal 2 ☐